

Frequency Comb Generation in Single-Mode Silicon Microresonators at 1550 nm in Normal Dispersion Regime

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Abstract We demonstrate frequency comb generation in a single-mode silicon microring resonator under normal dispersion at 1550 nm. Using a dual-pump configuration (2.5 mW and 27 μ W) and an 11 V bias, seven comb lines spanning 3.74 THz are observed, indicating stable and coherent oscillation.

Introduction

On-chip optical frequency combs play a vital role in several applications, including optical communication [1–3], high-capacity WDM systems [4], analog-to-digital converters (ADC) [5, 6], and microwave photonics applications [7–9]. On-chip comb generation is efficiently achieved in different platforms by leveraging nonlinear interactions within micro-ring resonators. Most demonstrations of microresonator-based frequency combs rely on anomalous dispersion, in a low-loss silicon nitride platform, to support soliton formation and achieve broadband spectra [10–12]. However, generating frequency combs on silicon photonics platforms is crucial and needs significant traction within the community due to CMOS compatibility, high index contrast, scalability, and low cost. Furthermore, for comb-based optical communications and ADC, a fully monolithic silicon-on-chip frequency comb source remains a critical milestone that has not yet been demonstrated. While silicon nitride platforms have achieved high-performance combs, they cannot be monolithically integrated with high-speed transceivers on a single substrate [1, 5]. In contrast, a silicon-based comb offers the unique advantage of direct integration with high-speed modulators on the same chip—a crucial architecture for next-generation silicon photonics interconnects that has yet to be fully realized.

To date, the realization of silicon frequency combs has been hindered by significant nonlinear losses, specifically two-photon absorption (TPA) and subsequent free-carrier absorption (FCA) at telecommunication wavelengths, in which the cavity is in an unstable regime [13, 14]. These effects diminish performance, but they do not, in themselves, prevent parametric gain as a seed for comb formation; a pulsed pump laser has been used to compensate for these losses and achieve net gain [15]. Standard silicon waveguides typically exhibit normal dispersion in this band, which severely restricts the achievable comb bandwidth.

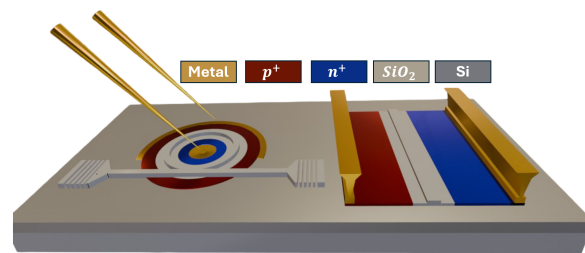


Fig. 1: Schematic illustration of an active silicon microring resonator probed by metal tips, with a cross-sectional view detailing the layered metal contacts and doped regions.

While recent work [16] demonstrated frequency comb formation in silicon at telecom wavelengths, that approach relied on millimeter-scale racetrack resonators with very wide waveguides to minimize loss, resulting in an efficient 1-FSR comb with a bandwidth limited to only 2 nm. This architecture faces significant scaling limitations; the large device footprint occupies excessive chip area and prevents high-density integration. Furthermore, that platform operated in a multimode regime hindered by intermodal dispersion, resulting in pulse spreading and strictly limited data bandwidth over distance. Additionally, these waveguides are prone to modal noise and unpredictable power fluctuations due to mode interference, making them less stable than single-mode alternatives. However, to the best of our knowledge, frequency comb generation in standard, small-footprint (radii of only a few micrometers), single-mode silicon microring resonators operating in the normal dispersion regime at telecommunication wavelengths has never been demonstrated. Generating frequency combs within small-footprint, single-mode silicon microring resonators is essential; by eliminating intermodal dispersion and mode-crossing interference, these waveguides ensure a predictable spectral output in the communication domain.

In this paper, we demonstrate frequency comb generation in an active 20 μ m-radius silicon microring resonator with a standard single-mode waveguide thickness of 220 nm and a width of 450 nm.

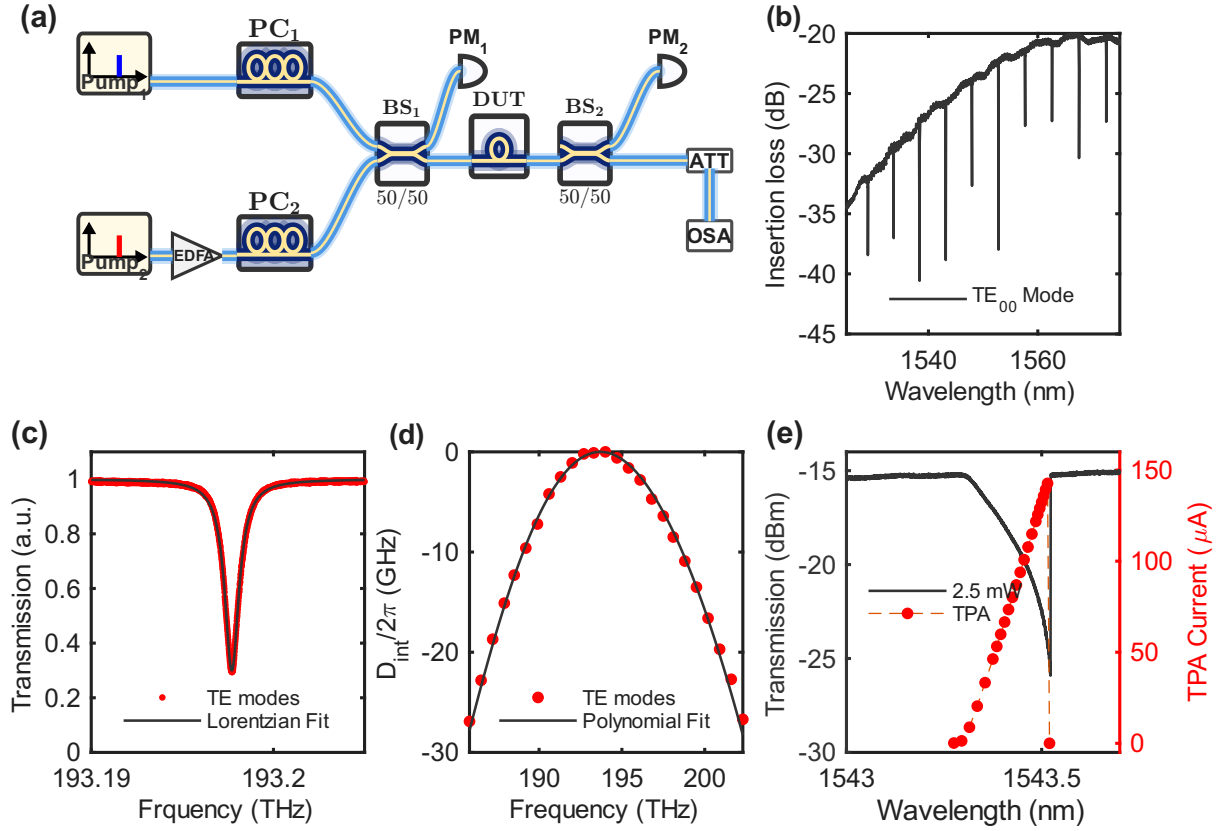


Fig. 2: Characterization of the integrated silicon microresonator. (a) Experimental setup; (b) transmission spectrum; (c) Lorentzian fit; (d) integral dispersion; (e) nonlinear transmission spectrum and corresponding absorption current.

Figure 1 shows the 3D silicon microcavity and its waveguide cross-section embedded in a PIN junction. This design features four doping concentrations and metal electrodes for free-carrier removal, which mitigates carrier-induced dispersion and absorption and ensures a stable parametric nonlinear process. By applying an 11 V reverse bias to reduce the free-carrier lifetime within the cavity [17], we generated a frequency comb consisting of seven lines. These lines are separated by 1 FSR and span approximately 30 nm around 1550 nm. These results highlight the possibility of achieving relatively broad spectral coverage in a CMOS-compatible platform without complex dispersion engineering, offering a simplified route toward fully monolithic integrated comb sources.

Experimental setup and characterization

Figure 2 (a) illustrates the experimental setup with the complete optical path used for device excitation and optical comb generation analysis. A continuous-wave laser weak pump (pump₁) is combined with a high-power pump (pump₂) laser and amplified using an erbium-doped fiber amplifier (EDFA). The two pumps are controlled by polarization controllers (PC₁ and PC₂) to optimize coupling into the desired fundamental transverse-electric (TE₀₀) mode. The guided lights are combined and routed through a 50:50 beam splitter (BS₁) as a coupler; one part is coupled into the

on-chip device via grating couplers, while the other is coupled into the power meter (PM₁) as a reference. The output light from the chip is fiber-coupled and split by BS₂; one part is connected to PM₂ for alignment and transmission measurement, while the other part is further attenuated using an optical attenuator (ATT) before being recorded by an optical spectrum analyzer (OSA). This dual-path configuration enables simultaneous linear and nonlinear characterization while maintaining precise control over optical pump power and electrical bias conditions.

The measured insertion loss of the TE₀₀ mode is plotted as a function of wavelength over the telecommunications C-band as depicted in Fig. 2 (b). The spectrum exhibits a gradual variation in baseline insertion loss (approximately -20 dB to -34 dB) due to grating-coupler losses as a function of wavelength, superimposed on a series of sharp resonant dips. These dips correspond to high-Q-factor modes supported by the microcavity. The regular spacing indicates a well-defined free spectral range (FSR) of 4.8 nm, while the depth and sharpness of the resonances confirm efficient coupling and low intrinsic loss. The overall spectral behavior demonstrates stable single-mode operation with minimal modal interference.

Figure 2 (c) presents a zoomed-in view of a representative resonance at 1550 nm in the frequency domain. The experimental data (red markers) are

well fitted by a Lorentzian function (solid black curve), confirming that the resonance is dominated by intrinsic cavity decay rate ($\kappa_{00}/2\pi = 729$ MHz) rather than external decay rate ($\kappa_{ex}/2\pi = 216$ MHz). The symmetric lineshape indicates negligible mode splitting or backscattering effects, suggesting high fabrication quality and minimal surface roughness. The extracted linewidth provides direct access to the loaded quality factor of 2.04×10^5 and photon lifetime of ~ 167 ps, which is a critical parameter for the power threshold for comb formation.

The integrated dispersion, $D_{\text{int}}/2\pi$, is plotted as a function of frequency in Fig. 2 (d). The experimental data (red circles) follow a smooth parabolic trend—the downward curvature indicating a normal dispersion regime—which is accurately captured by a polynomial fit (black curve). This behavior confirms that second-order dispersion ($D_2 < 0$) dominates the measured spectral range, while higher-order contributions remain negligible. Post-fitting analysis reveals critical device parameters: a fundamental free spectral range (FSR) of $D_1/2\pi \approx 600$ GHz and a second-order normal dispersion coefficient of $D_2/2\pi \approx -2.33$ GHz. Such a near-parabolic profile is essential for satisfying phase-matching conditions in nonlinear processes, including parametric oscillation and frequency comb generation. The ability to precisely engineer and characterize this dispersion highlights the high fidelity of the device design.

Figure 2 (e) displays on the primary axis (left) the nonlinear transmission spectrum of the high-power pump (2.5 mW) in the bus waveguide. The resonance reveals a pronounced distortion, due to optical bistability, and an extinction ratio of approximately 12 dB. Concurrently, the auxiliary axis (right) plots the measured TPA-induced photocurrent (red markers) as a function of wavelength while applying a 11 V reverse bias for free-carrier removal. The TPA current in the unbiased case (i.e., no applied voltage) was in the range of μA . While TPA causes slight performance degradation, but it does not inherently prevent comb formation. The strong correlation between resonance detuning and TPA current confirms the onset of nonlinear absorption within the cavity. Furthermore, the asymmetric resonance distortion indicates power-dependent refractive-index changes and absorption effects, consistent with nonlinear carrier dynamics. This simultaneous observation of optical transmission and electrical response provides direct evidence of efficient light–matter interaction within the device.

Comb generation in silicon microresonators

To achieve parametric gain, the foundation of frequency comb generation, within a silicon microcavity, the free-carrier lifetime (τ_c) must be at least

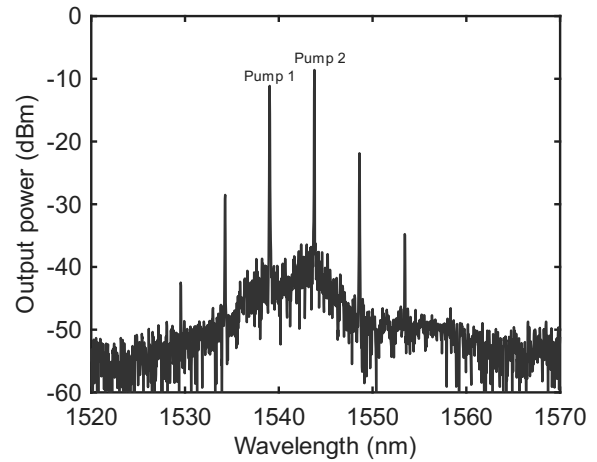


Fig. 3: Frequency comb spectrum from the integrated resonator. Under 11 V reverse bias, the Pump₁ and Pump₂ coupled powers are 27 μW and 2.5 mW, respectively.

50% lower than the photon lifetime (τ_{ph}) [18]. In our experiment, the most efficient comb generation occurs with a pump₁ power of 2.5 mW in the bus waveguide and a pump₂ power of 27 μW at an 11 V reverse bias applied to the PIN junction, achieving a ratio of $\tau_c/\tau_{ph} \approx 0.05$. Figure 3 illustrates the measured output optical spectrum as a function of wavelength, spanning approximately 1530–1560 nm. The spectrum exhibits a series of seven discrete, narrow spectral lines (comb teeth) separated by 1 FSR, which are superimposed on a broadband background envelope. The observed set of prominent peaks around the central wavelength region corresponds to the equally spaced resonant modes of the cavity, forming an optical frequency comb. The spacing between adjacent lines is exactly one FSR. These strong, well-defined peaks indicate efficient parametric gain and the coherent buildup of comb lines.

The spectral envelope displays a characteristic shape: the power is highest near the pump wavelength and gradually decreases toward the edges of the spectrum. This behavior is consistent with parametric four-wave mixing (FWM), in which energy is transferred from the pump to newly generated sidebands. The asymmetry and uneven peak amplitudes suggest the presence of dispersion and nonlinear loss mechanisms, which influence phase matching and comb efficiency across the bandwidth.

Conclusions

Silicon microresonators facilitate efficient frequency comb generation through precise dispersion engineering and carrier control. Utilizing dual pumps (2.5 mW and 27 μW) and an 11 V reverse bias, a 2.04×10^5 Q-factor cavity generates seven comb lines over ~ 30 nm. This monolithic silicon platform enables seamless integration with high-speed modulators, providing a scalable foundation for next-generation optical communications.

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